

1. [3 pts] Calcule $\overline{(i+3)} \cdot \frac{5i}{2-i}$.

$$(3-i) \cdot \frac{5i}{2-i} \cdot \frac{2+i}{2+i}$$

$$(3-i)(5i)(2+i)$$

$$9-i^2$$

$$(3-i) \frac{10i+5i^2}{9-1}$$

$$(3-i) \frac{10i-5}{8}$$

$$(3-i) \frac{5(2i-1)}{8}$$

$$(3-i)(2i-1)$$

$$6i-3-2i^2+i$$

$$7i-3-2$$

$$\boxed{7i-5}$$

2. Sean $x = 3 \operatorname{cis}(110^\circ)$, $y = 2 \operatorname{cis}(85^\circ)$, $z = xy$.

(a) [2 pts] Calcule z en forma rectangular.

$$z = 3 \operatorname{cis}(110^\circ) \cdot 2 \operatorname{cis}(85^\circ)$$

$$= 6 \operatorname{cis}(195^\circ)$$

$$6(\cos(195^\circ) + 6\sin(195^\circ)i)$$

$$\boxed{\frac{-3\sqrt{6}+3\sqrt{2}}{2} - \frac{3\sqrt{6}+3\sqrt{2}}{2}i}$$

(b) [1 pto] Encuentre el argumento principal de z .

$$z = \frac{-3\sqrt{6}-3\sqrt{2}}{2} - \frac{3\sqrt{6}+3\sqrt{2}}{2}i$$

$$\theta = \arctan\left(\frac{\frac{-3\sqrt{6}+3\sqrt{2}}{2}}{\frac{-3\sqrt{6}-3\sqrt{2}}{2}}\right) = \boxed{15}$$

3. [3 pts] Considere el polinomio $P(x) = x^5 - 3x^4 + 4x^3 - 4x^2 + 3x - 1$. Si se sabe que $P(i) = 0$, factorice completamente en $\mathbb{C}$ el polinomio $P(x)$.

$$P(i) = 0 \rightarrow x - i = 0 \rightarrow x = i$$

$$\begin{array}{r|rrrrrr} x^5 & -3x^4 & +4x^3 & -4x^2 & +3x & -1 \\ 1 & -3 & 4 & -4 & 3 & -1 \\ \hline & i & -3i-1 & 3+3i & -3-i & 1 \\ \hline 1 & -3+i & -3i+3 & -1+3i & -i & 0 \end{array} \quad (x-i)$$

$$\begin{array}{r|rrrrr} x^4 & +(-3+i)x^3 & +(-3i+3)x^2 & +(-1+3i)x & -i \\ 1 & -3+i & 3-3i & -1+3i & -i \\ \hline & -i & 3i & -3i & i \\ \hline 1 & -3 & 3 & -1 & 0 \end{array} \quad \begin{array}{l} (x-i) \\ (x+i) \end{array}$$

$$\begin{array}{r|rrrr} x^3 & -3x^2 & +3x & -1 \\ 1 & -3 & 3 & -1 \\ \hline & 1 & -2 & 1 \\ \hline 1 & -2 & 1 & 0 \end{array} \quad (x-1)$$

$$\begin{array}{r|rr} x^2 & -2x & +1 \\ 1 & -2 & 1 \\ \hline & 1 & -1 \\ \hline 1 & -1 & 0 \end{array} \quad (x-1)$$

$$(x-1)$$

$$(x-i)(x+i)(x-1)^3$$

4. [4 pts] Escriba en la forma $a + ib$ el número complejo $(2+i)^{1-i}$.

$$z = (2+i)^{1-i}$$

$$\ln(z) = (1-i) \ln(2+i)$$

$$= (1-i) \cdot [\ln(\sqrt{5}) + i \operatorname{arctan}(\frac{1}{2})]$$

$$= (1-i) [\ln(\sqrt{5}) + i \operatorname{arctan}(\frac{1}{2})]$$

$$\ln(\sqrt{5}) + i \operatorname{arctan}(\frac{1}{2}) - i \ln(\sqrt{5}) + \operatorname{arctan}(\frac{1}{2})$$

$$[\ln(\sqrt{5}) + i \operatorname{arctan}(\frac{1}{2})] + [-i \ln(\sqrt{5}) + \operatorname{arctan}(\frac{1}{2})]$$

$$e^{\ln(\sqrt{5}) + i \operatorname{arctan}(\frac{1}{2})} \cdot e^{-i \ln(\sqrt{5}) + \operatorname{arctan}(\frac{1}{2})}$$

$$e^{\ln(\sqrt{5}) + i \operatorname{arctan}(\frac{1}{2})} \cdot [(\cos(-\ln(\sqrt{5})) + i \sin(-\ln(\sqrt{5}))) + (\cos(\operatorname{arctan}(\frac{1}{2})) + i \sin(\operatorname{arctan}(\frac{1}{2})))]$$

$$3.350259375 - 1.789750222i$$

$$z = 2+i$$

$$a=2$$

$$b=1$$

$$r = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$\theta = \operatorname{arctan}(\frac{1}{2})$$

$$z = \sqrt{5} \cdot (i)^{\operatorname{arctan}(\frac{1}{2})}$$

$$z = \sqrt{5} \cdot e^{i \operatorname{arctan}(\frac{1}{2})}$$

5. [3 pts] Sea a una constante real no nula y sean

$$A = \begin{pmatrix} 1 & -1 \\ a+3 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 & a \\ 2 & -1 & 0 \end{pmatrix} \text{ y } C = \begin{pmatrix} 0 & 3 \\ -2 & 0 \\ 0 & -1 \end{pmatrix}$$

Compruebe que $A + \frac{1}{2}(BC)^T = I_2$.

$$BC = \begin{pmatrix} -1 & 0 & a \\ 2 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 3 \\ -2 & 0 \\ 0 & -1 \end{pmatrix}$$

$$BC = \begin{pmatrix} 0+0+0 & -3+0-a \\ 0+2+0 & 6+0+0 \end{pmatrix}$$

$$BC = \begin{pmatrix} 0 & -3-a \\ 2 & 6 \end{pmatrix}$$

$$(BC)^T = \begin{pmatrix} 0 & 2 \\ -3-a & 6 \end{pmatrix}$$

$$\frac{1}{2}(BC)^T = \begin{pmatrix} 0 & 1 \\ -\frac{3+a}{2} & 3 \end{pmatrix}$$

$$A + \frac{1}{2} (B C)^T = \begin{pmatrix} 0 & 1 \\ \frac{-3-9}{2} & 3 \end{pmatrix}$$

$$A + \frac{1}{2} (B C)^T = \begin{pmatrix} 1 & -1 \\ \frac{9+3}{2} & -2 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ \frac{-3-9}{2} & 3 \end{pmatrix}$$

$$A + \frac{1}{2} (B C)^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \neq I_2$$

6. [3 pts] Considere matrices A, B y C de orden $n \times n$ y $A^T - 2B$ invertible, tales que $(AX^T)^T = C^T + 2(XB)$. Si se sabe que C es una matriz simétrica, use las operaciones con matrices y sus propiedades para demostrar que $X = C(A^T - 2B)^{-1}$.

$$A^T - 2B \rightarrow (A^T - 2B)^{-1}$$

$$C^T = C$$

$$(AX^T)^T = C^T + 2XB$$

$$XA^T - 2XB = C^T$$

$$X(A^T - 2B) = C$$

$$X = \frac{C}{A^T - 2B}$$

$$X = C \cdot (A^T - 2B)^{-1}$$

7. [5 pts] Use el método de Gauss-Jordan para determinar el conjunto solución del sistema:

$$\begin{cases} x + y + z = 6 \\ 2x - y + 3z = 14 \\ 2x + 2y + 2z = 12 \end{cases}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 3 & 14 \\ 2 & 2 & 2 & 12 \end{array} \right)$$

$$\begin{array}{l} -2 \cdot F_1 + \widetilde{F}_2 \\ -2 \cdot F_1 + \widetilde{F}_3 \end{array} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\begin{array}{l} -F_2 + \widetilde{F}_1 \\ -\frac{1}{3}, \widetilde{F}_2 \end{array} \left(\begin{array}{ccc|c} 1 & 0 & \frac{7}{3} & \frac{20}{3} \\ 0 & 1 & -\frac{1}{3} & -\frac{2}{3} \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\begin{cases} x + \frac{7}{3}z = \frac{20}{3} & \rightarrow x = \frac{20}{3} - \frac{7}{3}z \\ y - \frac{1}{3}z = -\frac{2}{3} & y = -\frac{2}{3} + \frac{1}{3}z \end{cases}$$

$$S = \left\{ \left(\frac{20}{3} - \frac{7}{3}z, -\frac{2}{3} + \frac{1}{3}z, z \right), z \in \mathbb{R} \right\}$$

8. [4 pts] Considere la matriz $A = \begin{pmatrix} -7 & 5 & 4 \\ 2 & -1 & -1 \\ 4 & -3 & -2 \end{pmatrix}$ y determine A^{-1} .

$$\begin{array}{l} -\frac{1}{7}, \widetilde{F}_1 \\ -2, F_1 + \widetilde{F}_2 \\ -4, F_1 + \widetilde{F}_3 \end{array} \left(\begin{array}{ccc|ccc} 1 & -\frac{5}{7} & -\frac{4}{7} & -\frac{1}{7} & 0 & 0 \\ 0 & \frac{3}{7} & \frac{1}{7} & \frac{2}{7} & 1 & 0 \\ 0 & -\frac{1}{7} & \frac{2}{7} & \frac{4}{7} & 0 & 1 \end{array} \right)$$

$$\begin{array}{l} \frac{5}{7}, F_2 + \widetilde{F}_1 \\ \frac{1}{3}, \widetilde{F}_2 \\ \frac{1}{7}, F_2 + \widetilde{F}_3 \end{array} \left(\begin{array}{ccc|ccc} 1 & 0 & -\frac{1}{3} & \frac{1}{3} & \frac{5}{3} & 0 \\ 0 & 1 & \frac{1}{3} & \frac{2}{3} & \frac{7}{3} & 0 \\ 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 1 \end{array} \right)$$

$$\begin{array}{l} \frac{1}{3}, F_3 + \widetilde{F}_1 \\ -\frac{1}{3}, F_3 + \widetilde{F}_2 \\ 3, \widetilde{F}_3 \end{array} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & 1 \\ 0 & 1 & 0 & 0 & 2 & -1 \\ 0 & 0 & 1 & 2 & 1 & 3 \end{array} \right)$$

$$A^{-1} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 2 & 1 & 3 \end{pmatrix}$$